

**RV College of Engineering[®] ,
(Autonomous Institution affiliated to VTU)
Bengaluru-560 059**

**Course Title: Mathematical Algorithms for Artificial Intelligence
Model Question Paper**

COURSE CODE: AI255TBC	SEM: V
COURSE TITLE : Mathematical Algorithms for AI	
Duration of Paper: 03 Hrs	

Instructions to Candidates:

1. Answer all questions from Part A
2. Any 5 Full questions from Part B choosing one from each side. (Question No.2 is compulsory)

Q.No	PART A	Marks	BTL	CO
1.1	If the column space of an 8×4 matrix A is 3 dimensional, give the dimensions of the other three fundamental subspaces.	02		
1.2	Assert or Reject the following statement with justification: Statement: The set of all even degree polynomials (including 0) is a vector space.	02		
1.3	Given the Euclidean norms $\ u\ = 4$, $\ v\ = 3$, $\ u + v\ = 7$, then what is the dot product $\langle u, v \rangle$?	02		
1.4	Assert or Reject the following statement with justification: Statement: There exists a real symmetric matrix whose characteristic equation is $\lambda(\lambda^2 + 4)(\lambda - 1) = 0$.	02		
1.5	Define heteroscedasticity.	02		
1.6	How is the ordinary least squares problem different from a weighted least squares case?	02		
1.7	The position vector of point P is $r = (3 \cos t, 5 \sin t, 4 \cos t)$. What is the speed?	02		
1.8	Obtain the gradient of the function $f(x, y) = (x + y)^2$	02		
1.9	Consider a 2×3 real matrix A, with rank of A = 2. What is the pseudoinverse of A?	02		
1.10	What could be the consequence of higher learning rate in gradient descent algorithm?	02		
PART B				
2(a)	Let $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ Find out all the 4 subspaces associated with A. Mention one basis each for each of the subspaces.	10		
2(b)	Assert or Reject the following statements with justification.	6		

	Statement 1: For a given $n \times n$ real matrix A, two eigenvalues can have the same eigenvectors. Statement 2: $\mathbb{R}^2$ is a subspace of $\mathbb{R}^3$.			
3(a)	Given the following vectors $\mathbf{u}_1 = \begin{pmatrix} 3 \\ -3 \\ 0 \end{pmatrix}, \mathbf{u}_2 = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}, \mathbf{u}_3 = \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} 5 \\ -3 \\ 1 \end{pmatrix}$ (i) Are the vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ orthogonal? (ii) Are the vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$ orthonormal? (iii) Express the vector $\mathbf{x}$ as a linear combination of the vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3$.	8 2+1+5		
3(b)	(i) If P is a projection matrix, prove that $P^2 = P$. (ii) Obtain the matrix that projects every vector in $\mathbb{R}^2$ along the line passing through the origin and in the direction of the vector $\mathbf{u}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$	4+4		
OR				
4(a)	Obtain the orthogonal complement subspace of the plane $x_1 + x_2 + x_3 = 0$	6		
4(b)	Consider the following data matrix: $X = \begin{pmatrix} 4 & 1 \\ 2 & 3 \\ 5 & 4 \\ 1 & 0 \end{pmatrix}$ We want to represent the data in only one dimension. Compute the unit-length principal component directions of X, and state which one the PCA algorithm would choose if you request just one principal component. Obtain the projections of all four sample points onto the principal direction.	10		
5(a)	Suppose you measure your pulse rate x three times and get values $x = b_1, x = b_2, x = b_3$. If each right side b_i , are independent and has zero mean and variances $\sigma_1^2 = \frac{1}{16}, \sigma_2^2 = \frac{1}{9}, \sigma_3^2 = \frac{1}{36}$, find the best estimate $\hat{x}$ based on b_1, b_2, b_3 .	10		
5(b)	List out 3 limitations of Gaussian Mixture Models	6		
OR				
6(a)	List out 4 limitations of Expectation Maximization algorithm	12		
6(b)	List 4 applications of Kalman filters	4		
7(a)	The quadratic form $\mathbf{x}^T A \mathbf{x}$ is a form we will encounter often.* In this question, we are interested in $d(\mathbf{x}^T A \mathbf{x})/d\mathbf{x}$. Assume that A is not a function of $\mathbf{x}$. (a) Evaluate $\mathbf{x}^T A \mathbf{x}$ when $\mathbf{x} = [x_1, x_2]^T$ and the (i,j)-th element of A is A_{ij}. Why do you think $\mathbf{x}^T A \mathbf{x}$ is called the quadratic form?	16		

	(b) Which definition of the derivative do we need in order to evaluate $d(\mathbf{x}^T A \mathbf{x})/d\mathbf{x}$? (c) Assume $\mathbf{x} \in \mathbb{R}^2$ and $A \in \mathbb{R}^{2 \times 2}$. Evaluate $d(\mathbf{x}^T A \mathbf{x})/d\mathbf{x}$. (d) Generalize the previous result to when $\mathbf{x} \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$ and evaluate $d(\mathbf{x}^T A \mathbf{x})/d\mathbf{x}$. Can you express the result in matrix form? (e) What happens when A is a symmetric matrix, i.e., $A^T = A$?			
	OR			
8	Suppose the 2x1 function $\mathbf{f}$ of the 3x1 vector $\mathbf{x}$ is given by $\mathbf{f}(\mathbf{x}) = \begin{pmatrix} x_1^2 + x_2^2 + x_3^2 \\ 2x_1 - x_2 - x_3 \end{pmatrix}$ and the 2x1 function $\mathbf{g}$ of the 2x1 vector $\mathbf{z}$ is given by $\mathbf{g}(\mathbf{z}) = \begin{pmatrix} z_2 \\ z_1 \\ z_1 z_2 \end{pmatrix}$ Use the chain rule to compute $\frac{\partial}{\partial \mathbf{x}^T} \mathbf{y}(\mathbf{x})$ where $\mathbf{y}(\mathbf{x})$ is the composite function defined by $\mathbf{y}(\mathbf{x}) = \mathbf{g}(\mathbf{f}(\mathbf{x}))$.	16		
9(a)	The function $f(x, y, z) = x^2 + y^2 + 3z^2 - xy + 2xz + yz$ defined on $\mathbb{R}^3$ has only one stationary point. Show that it is a local minimum.	6		
9(b)	Find the least square solution to the following system of equations: $x_1 + x_2 = 2; x_1 - x_2 = 0; 2x_1 + x_2 = -4$	10		
	OR			
10(a)	Use the method of Lagrange multipliers to find the minimum value of $x^2 + 4y^2 - 2x + 8y$ subject to the constraint $x+2y=7$	16		

Name of the Scrutinizer

Name of the BoE Chairperson

Signature of Scrutinizer

Name of the BoE Chairperson

Dimensions of Fundamental Subspaces

Problem: If the Column Space of an 8×4 matrix A is 3-dimensional, give the dimensions of the other three fundamental subspaces.

Given:

- Matrix A has size 8×4 .
 - $m = 8$ (Number of Rows / Output Dimension)
 - $n = 4$ (Number of Columns / Input Dimension)
- $\dim(C(A)) = 3$.

1. Identify the Rank

The dimension of the Column Space is defined as the **Rank** (r) of the matrix.

$$r = 3$$

2. The Four Fundamental Subspaces

We use the Fundamental Theorem of Linear Algebra to find the remaining dimensions.

A. Column Space $C(A)$

- **Definition:** Range of A . Lives in $\mathbb{R}^m$.
- **Dimension:** Equal to the Rank.

$$\dim(C(A)) = r = 3$$

B. Row Space $C(A^T)$

- **Definition:** Range of A^T . Lives in $\mathbb{R}^n$.
- **Theorem:** Row Rank equals Column Rank.

$$\dim(C(A^T)) = r = 3$$

C. Null Space $N(A)$

- **Definition:** x such that $Ax = 0$. Lives in $\mathbb{R}^n$ (Input).
- **Theorem:** Rank + Nullity = n (Total Columns).

$$r + \dim(N(A)) = 4$$

$$3 + \dim(N(A)) = 4$$

$$\dim(N(A)) = 1$$

D. Left Null Space $N(A^T)$

- **Definition:** y such that $A^T y = 0$. Lives in $\mathbb{R}^m$ (Output).
- **Theorem:** Rank + Left Nullity = m (Total Rows).

$$r + \dim(N(A^T)) = 8$$

$$3 + \dim(N(A^T)) = 8$$

$$\dim(N(A^T)) = 5$$

Final Answer Summary

Subspace	Symbol	Dimension Formula	Result
Column Space	$C(A)$	r	3
Row Space	$C(A^T)$	r	3
Null Space	$N(A)$	$n - r$	1
Left Null Space	$N(A^T)$	$m - r$	5

Solutions to Linear Algebra, Calculus, and ML Queries

2. Vector Space of Even Degree Polynomials

Statement: The set of all even degree polynomials including 0 is a vector space.

Verdict: Reject. (The statement is False).

Justification: For a set to be a vector space, it must be closed under vector addition. Let V be the set of polynomials with even degree (plus the zero polynomial). Consider two polynomials in this set:

1. $P(x) = x^2 + x$ (Degree is 2, which is even).
2. $Q(x) = -x^2$ (Degree is 2, which is even).

Now, add them together:

$$P(x) + Q(x) = (x^2 + x) + (-x^2) = x$$

The resulting polynomial is x , which has degree 1 (odd). Since the result lies outside the set, the set is not closed under addition and is therefore **not a vector space**.

3. Dot Product from Norms

Problem: Given $\|u\| = 4$, $\|v\| = 3$, and $\|u + v\| = 7$. Find $\langle u, v \rangle$.

Solution: We use the expansion of the squared norm:

$$\|u + v\|^2 = \langle u + v, u + v \rangle = \|u\|^2 + \|v\|^2 + 2\langle u, v \rangle$$

Substitute the given values:

$$7^2 = 4^2 + 3^2 + 2\langle u, v \rangle$$

$$49 = 16 + 9 + 2\langle u, v \rangle$$

$$49 = 25 + 2\langle u, v \rangle$$

$$24 = 2\langle u, v \rangle$$

$$\langle u, v \rangle = 12$$

(Note: Since $\|u\| + \|v\| = \|u + v\|$, this represents the equality case of the Triangle Inequality, implying the vectors are parallel and point in the same direction.)

4. Real Symmetric Matrix Characteristic Equation

Statement: There exists a real symmetric matrix whose characteristic equation is $\lambda(\lambda^2 + 4)(\lambda - 1) = 0$.

Verdict: Reject. (The statement is False).

Justification: Solving the characteristic equation for the eigenvalues (λ):

$$\lambda(\lambda^2 + 4)(\lambda - 1) = 0$$

The roots are:

- 1. $\lambda_1 = 0$
- 2. $\lambda_2 = 1$
- 3. $\lambda^2 + 4 = 0 \implies \lambda_{3,4} = \pm 2i$ (Complex Imaginary numbers)

Spectral Theorem for Real Symmetric Matrices: All eigenvalues of a real symmetric matrix **must be real**. Since the proposed matrix requires complex eigenvalues ($2i, -2i$), it **cannot** be a real symmetric matrix.

5. Define Heteroscedasticity

Definition: In statistics and regression analysis, **heteroscedasticity** refers to the condition where the variance of the residuals (errors) is **not constant** across all levels of the independent variable(s).

Visually, in a scatter plot of residuals vs. predicted values, heteroscedasticity often looks like a cone or a fan shape (the spread of errors gets wider or narrower as the value of x changes). This violates the standard OLS assumption of homoscedasticity (constant variance).

6. Ordinary Least Squares (OLS) vs. Weighted Least Squares (WLS)

Feature	Ordinary Least Squares (OLS)	Weighted Least Squares (WLS)
Assumption	Assumes Homoscedasticity (All error terms have the same variance, σ^2).	Handles Heteroscedasticity (Error terms have different variances, σ_i^2).
Objective	Minimize the sum of squared residuals: $\sum (y_i - \hat{y}_i)^2$	Minimize the weighted sum of squared residuals: $\sum w_i (y_i - \hat{y}_i)^2$
Weighting	All observations are treated equally (Weight = 1).	Observations with lower variance (higher precision) are given higher weights ($w_i \propto 1/\sigma_i^2$).

7. Speed of a Position Vector

Problem: Position vector $\mathbf{r}(t) = (3 \cos t, 5 \sin t, 4 \cos t)$. What is the speed?

Solution:

1. **Find Velocity ($\mathbf{v}$):** Differentiate position with respect to t .

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = (-3 \sin t, 5 \cos t, -4 \sin t)$$

2. **Find Speed ($\|\mathbf{v}\|$):** Calculate the magnitude of velocity.

$$\|\mathbf{v}\| = \sqrt{(-3 \sin t)^2 + (5 \cos t)^2 + (-4 \sin t)^2}$$

$$\|\mathbf{v}\| = \sqrt{9 \sin^2 t + 25 \cos^2 t + 16 \sin^2 t}$$

Combine the $\sin^2 t$ terms:

$$\|\mathbf{v}\| = \sqrt{25 \sin^2 t + 25 \cos^2 t}$$

$$\|\mathbf{v}\| = \sqrt{25(\sin^2 t + \cos^2 t)}$$

$$\|\mathbf{v}\| = \sqrt{25(1)} = 5$$

Answer: The speed is constant at 5.

8. Gradient of $f(x,y)$

Problem: Obtain the gradient of $f(x,y) = (x+y)^2$.

Solution: The gradient ∇f is the vector of partial derivatives.

$$\nabla f = \begin{bmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \end{bmatrix}$$

$$1. \quad \frac{\partial}{\partial x}(x+y)^2 = 2(x+y) \cdot (1) = 2(x+y)$$

$$2. \quad \frac{\partial}{\partial y}(x+y)^2 = 2(x+y) \cdot (1) = 2(x+y)$$

Answer:

$$\nabla f = \begin{bmatrix} 2(x+y) \\ 2(x+y) \end{bmatrix}$$

9. Pseudoinverse of a 2x3 Matrix

Problem: Consider a 2×3 real matrix A with Rank = 2. What is the pseudoinverse?

Solution: Since A is 2×3 and has Rank 2, it has **Full Row Rank** ($m < n$, $\text{rank} = m$). For a matrix with full row rank, the rows are linearly independent, which means the matrix AA^T is invertible.

The pseudoinverse ($A^\dagger$ or A^+) is given by the **Right Inverse** formula:

$$A^\dagger = A^T(AA^T)^{-1}$$

(Note: If the matrix had Full Column Rank, the formula would be the Left Inverse: $(A^T A)^{-1} A^T$.)

10. Consequence of High Learning Rate

Question: What could be the consequence of a higher learning rate in the gradient descent algorithm?

Answer: A learning rate that is set too high can lead to:

1. **Overshooting:** The algorithm takes steps that are too large, jumping over the global minimum.
2. **Divergence:** Instead of minimizing the loss function, the parameters might spiral out of control, causing the loss to increase towards infinity (exploding gradients).
3. **Oscillation:** The algorithm may bounce back and forth around the minimum without ever

Linear Algebra: Subspaces & Assertions

1. Four Fundamental Subspaces of Matrix A

Problem: Let A be a 3×2 matrix:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$$

Find out all the 4 subspaces associated with A . Mention one basis each for each of the subspaces with reference to Gilbert Strang's diagram of four fundamental subspaces of a matrix.

Analysis of Matrix A:

- **Dimensions:** $m = 3$ (Rows), $n = 2$ (Columns).
- **Input Space:** Vectors x live in $\mathbb{R}^n = \mathbb{R}^2$.
- **Output Space:** Vectors b live in $\mathbb{R}^m = \mathbb{R}^3$.
- **Rank:** The two columns are clearly linearly independent (not multiples of each other). Thus, Rank $r = 2$.

A. Subspaces in the Input Space ($\mathbb{R}^2$)

According to Strang's diagram, the Row Space and Null Space reside in $\mathbb{R}^2$ and are **orthogonal complements**. Their dimensions sum to $n = 2$.

1. Row Space ($C(A^T)$)

- **Definition:** The subspace spanned by the rows of A .
- **Analysis:** The rows are $r_1 = (1, 0)$, $r_2 = (0, 1)$, and $r_3 = (1, 1)$. Notice that $r_3 = r_1 + r_2$. Therefore, the dimension is 2.
- **Dimension:** $r = 2$.
- **Basis:** The first two rows form the simplest basis (standard basis for $\mathbb{R}^2$).

$$\text{Basis} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

2. Null Space ($N(A)$)

- **Definition:** The set of all vectors x such that $Ax = 0$.
- **Calculation:**

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

This system gives the equations:

1. $x_1 = 0$
 2. $x_2 = 0$
 3. $x_1 + x_2 = 0$ The only vector satisfying this is the zero vector.
- **Dimension:** $n - r = 2 - 2 = 0$.
 - **Basis:** The Null Space contains only the zero vector $\{\vec{0}\}$. By convention, the basis is the empty set $\emptyset$.

B. Subspaces in the Output Space ($\mathbb{R}^3$)

According to Strang's diagram, the Column Space and Left Null Space reside in $\mathbb{R}^3$ and are **orthogonal complements**. Their dimensions sum to $m = 3$.

3. Column Space ($C(A)$)

- **Definition:** The subspace spanned by the columns of A .
- **Analysis:** Since the rank is 2, the columns are linearly independent. They form a plane passing through the origin in 3D space.
- **Dimension:** $r = 2$.
- **Basis:** The columns themselves form a basis.

$$\text{Basis} = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$$

4. Left Null Space ($N(A^T)$)

- **Definition:** The set of all vectors y such that $A^T y = 0$. (Vectors orthogonal to the columns).
- **Calculation:**

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

This gives the equations:

1. $y_1 + y_3 = 0 \implies y_1 = -y_3$
2. $y_2 + y_3 = 0 \implies y_2 = -y_3$

Let y_3 be a free variable. If we set $y_3 = 1$, then $y_1 = -1$ and $y_2 = -1$.

- **Dimension:** $m - r = 3 - 2 = 1$ (A line in $\mathbb{R}^3$).
- **Basis:**

$$\text{Basis} = \left\{ \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} \right\}$$

Verification (Strang's Diagram): Check if the Column Space basis is orthogonal to the Left Null Space basis:

$$\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} = -1 + 0 + 1 = 0 \quad (\text{Orthogonal})$$

2. Assertions with Justification

Statement 1: "For a given $n \times n$ real matrix A , two eigenvalues can have the same eigenvectors."

Verdict: Reject.

Justification: By definition, an eigenvector $\mathbf{x}$ must be a non-zero vector. Let's assume a single vector $\mathbf{x} \neq \vec{0}$ is an eigenvector for two **distinct** eigenvalues λ_1 and λ_2 (where $\lambda_1 \neq \lambda_2$).

1. By definition: $A\mathbf{x} = \lambda_1\mathbf{x}$
2. By definition: $A\mathbf{x} = \lambda_2\mathbf{x}$

Subtracting equation (2) from (1):

$$\lambda_1\mathbf{x} - \lambda_2\mathbf{x} = \vec{0}$$

$$(\lambda_1 - \lambda_2)\mathbf{x} = \vec{0}$$

Since $\mathbf{x} \neq \vec{0}$, the scalar $(\lambda_1 - \lambda_2)$ must be zero. This implies $\lambda_1 = \lambda_2$. Therefore, it is impossible for two **distinct** eigenvalues to share the same eigenvector. The statement is false.

(Note: If the eigenvalues are identical, i.e., algebraic multiplicity > 1 , they naturally share eigenvectors, but in linear algebra context, "two eigenvalues" typically implies distinct values.)

Statement 2: " $\mathbb{R}^2$ is a subspace of $\mathbb{R}^3$."

Verdict: Reject.

Justification: For a set W to be a subspace of a vector space V , W must be a **subset** of V ($W \subseteq V$).

- Elements of $\mathbb{R}^2$ are ordered pairs, e.g., $\mathbf{u} = (x, y)$.
- Elements of $\mathbb{R}^3$ are ordered triples, e.g., $\mathbf{v} = (x, y, z)$.

Mathematically, an ordered pair is not the same object as an ordered triple. The set of pairs is not contained within the set of triples. Therefore, $\mathbb{R}^2$ is strictly **not** a subspace of $\mathbb{R}^3$.

Nuance: The set of vectors in $\mathbb{R}^3$ where the third component is zero,

$W = \{(x, y, 0) \mid x, y \in \mathbb{R}\}$, is a subspace of $\mathbb{R}^3$. While W is **isomorphic** to $\mathbb{R}^2$ (they behave identically), they are not the same set. The statement as written is false.

3. Orthogonality and Basis Expansion

Problem: Given the following 3×1 vectors:

$$u_1 = (3, -3, 0), \quad u_2 = (2, 2, -1), \quad u_3 = (1, 1, 4)$$

And vector $x = (5, -3, 1)$.

i) Are the vectors u_1, u_2, u_3 orthogonal?

To check for orthogonality, we compute the dot product for every distinct pair. If the dot product is zero, the vectors are orthogonal.

1. $u_1 \cdot u_2 = (3)(2) + (-3)(2) + (0)(-1) = 6 - 6 + 0 = 0.$
2. $u_1 \cdot u_3 = (3)(1) + (-3)(1) + (0)(4) = 3 - 3 + 0 = 0.$
3. $u_2 \cdot u_3 = (2)(1) + (2)(1) + (-1)(4) = 2 + 2 - 4 = 0.$

Answer: Yes, the vectors are mutually orthogonal.

ii) Are the vectors u_1, u_2, u_3 orthonormal?

A set of vectors is **orthonormal** if they are orthogonal AND each vector has a magnitude (norm) of 1.

We calculate the squared norms ($\|u\|^2 = u \cdot u$):

1. $\|u_1\|^2 = 3^2 + (-3)^2 + 0^2 = 9 + 9 + 0 = 18.$
 - Magnitude: $\sqrt{18} \neq 1.$
2. $\|u_2\|^2 = 2^2 + 2^2 + (-1)^2 = 4 + 4 + 1 = 9.$
 - Magnitude: $\sqrt{9} = 3 \neq 1.$
3. $\|u_3\|^2 = 1^2 + 1^2 + 4^2 = 1 + 1 + 16 = 18.$
 - Magnitude: $\sqrt{18} \neq 1.$

Answer: No, the vectors are **not** orthonormal because they are not unit vectors.

iii) Express the vector x as a linear combination of the vectors u_1, u_2, u_3 .

We want to find scalars c_1, c_2, c_3 such that:

$$x = c_1 u_1 + c_2 u_2 + c_3 u_3$$

Since the basis $\{u_1, u_2, u_3\}$ is **orthogonal**, we can find the coefficients using the projection formula (Fourier coefficients):

$$c_i = \frac{x \cdot u_i}{u_i \cdot u_i} = \frac{x \cdot u_i}{||u_i||^2}$$

Calculate c_1 :

- $x \cdot u_1 = (5)(3) + (-3)(-3) + (1)(0) = 15 + 9 + 0 = 24$
- $||u_1||^2 = 18$
- $c_1 = \frac{24}{18} = \frac{4}{3}$

Calculate c_2 :

- $x \cdot u_2 = (5)(2) + (-3)(2) + (1)(-1) = 10 - 6 - 1 = 3$
- $||u_2||^2 = 9$
- $c_2 = \frac{3}{9} = \frac{1}{3}$

Calculate c_3 :

- $x \cdot u_3 = (5)(1) + (-3)(1) + (1)(4) = 5 - 3 + 4 = 6$
- $||u_3||^2 = 18$
- $c_3 = \frac{6}{18} = \frac{1}{3}$

Final Expression:

$$x = \frac{4}{3}u_1 + \frac{1}{3}u_2 + \frac{1}{3}u_3$$

4. Projection Matrices

i) If P is a projection matrix, prove that $P^2 = P$.

Proof: A projection matrix P projects any vector $\mathbf{b}$ onto a subspace. Let $\mathbf{p} = P\mathbf{b}$ be the projection. Since $\mathbf{p}$ is already in the subspace, projecting it again should not change it. Geometrically, this means $P\mathbf{p} = \mathbf{p}$.

Substituting $\mathbf{p} = P\mathbf{b}$:

$$P(P\mathbf{b}) = P\mathbf{b}$$

$$P^2\mathbf{b} = P\mathbf{b}$$

Since this holds for any arbitrary vector $\mathbf{b}$, the matrices must be equal:

$$P^2 = P$$

Algebraic Proof (using the Orthogonal Projection Formula): Let $P = A(A^T A)^{-1} A^T$, where the columns of A form a basis for the subspace.

$$\begin{aligned}
 P^2 &= (A(A^T A)^{-1} A^T)(A(A^T A)^{-1} A^T) \\
 &= A(A^T A)^{-1} \underbrace{(A^T A)(A^T A)^{-1}}_I A^T \\
 &= A(A^T A)^{-1} I A^T \\
 &= A(A^T A)^{-1} A^T \\
 &= P
 \end{aligned}$$

Thus, P is **idempotent**.

ii) Obtain the matrix that projects every vector in $\mathbb{R}^2$ along the line passing through the origin and in the direction of the vector $u_1 = (1, 1)$.

Solution: The formula for the projection matrix P onto the line spanned by a column vector $\mathbf{a}$ is:

$$P = \frac{\mathbf{a}\mathbf{a}^T}{\mathbf{a}^T \mathbf{a}}$$

Here, our direction vector is $\mathbf{a} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

1. **Compute the Outer Product ($\mathbf{a}\mathbf{a}^T$):**

$$\mathbf{a}\mathbf{a}^T = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

2. **Compute the Inner Product / Squared Norm ($\mathbf{a}^T \mathbf{a}$):**

$$\mathbf{a}^T \mathbf{a} = \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = 1^2 + 1^2 = 2$$

3. **Construct the Matrix:**

$$P = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}$$

Answer:

$$P = \begin{pmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{pmatrix}$$

5. Orthogonal Complement of a Plane

Problem: Obtain the orthogonal complement subspace of the plane defined by

$$x_1 + x_2 + x_3 = 0.$$

Solution:

1. **Identify the Subspace W :** The plane is the set of all vectors $\mathbf{x} = (x_1, x_2, x_3)$ that satisfy the linear equation:

$$1 \cdot x_1 + 1 \cdot x_2 + 1 \cdot x_3 = 0$$

This equation can be rewritten as a dot product between a specific vector $\mathbf{n}$ and any vector $\mathbf{x}$ in the plane:

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

Here, $\mathbf{n} = (1, 1, 1)$ is the **normal vector** to the plane.

2. **Definition of Orthogonal Complement ($W^\perp$):** $W^\perp$ consists of all vectors that are orthogonal to every vector in W . From step 1, we see that the vector $\mathbf{n} = (1, 1, 1)$ is orthogonal to every $\mathbf{x}$ in the plane. Therefore, $\mathbf{n} \in W^\perp$.
3. **Check Dimensions:**
 - The ambient space is $\mathbb{R}^3$ ($n = 3$).
 - The subspace W is a plane, so its dimension is 2.
 - By the property of orthogonal complements: $\dim(W) + \dim(W^\perp) = 3$.
 - Therefore, $\dim(W^\perp) = 3 - 2 = 1$.
4. **Conclusion:** Since $W^\perp$ is 1-dimensional (a line) and we found a non-zero vector $\mathbf{n} = (1, 1, 1)$ inside it, this vector must form the basis for $W^\perp$.

Answer: The orthogonal complement is the line spanned by the vector $(1, 1, 1)$.

$$W^\perp = \text{span} \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

6. Principal Component Analysis (PCA)

Problem: Consider the data matrix:

$$X = \begin{pmatrix} 4 & 1 \\ 2 & 3 \\ 5 & 4 \\ 1 & 0 \end{pmatrix}$$

We want to represent the data in one dimension. Compute the unit length principal component directions, choose the best one, and obtain the projections of the sample points.

Solution:

Step 1: Center the Data (Subtract the Mean) First, calculate the mean of each column (feature).

- Mean of Col 1 (x): $\bar{x} = \frac{4+2+5+1}{4} = \frac{12}{4} = 3$
- Mean of Col 2 (y): $\bar{y} = \frac{1+3+4+0}{4} = \frac{8}{4} = 2$
- Mean Vector: $\mu = (3, 2)$

Subtract μ from each row of X to get the centered matrix B :

$$B = \begin{pmatrix} 4-3 & 1-2 \\ 2-3 & 3-2 \\ 5-3 & 4-2 \\ 1-3 & 0-2 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \\ 2 & 2 \\ -2 & -2 \end{pmatrix}$$

Step 2: Compute the Covariance Matrix (Scatter Matrix) We compute $S = B^T B$. (Note: The actual covariance is $\frac{1}{N-1} B^T B$, but the scaling factor doesn't change the eigenvectors).

$$S = \begin{pmatrix} 1 & -1 & 2 & -2 \\ -1 & 1 & 2 & -2 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 1 \\ 2 & 2 \\ -2 & -2 \end{pmatrix}$$

- $S_{11} = 1^2 + (-1)^2 + 2^2 + (-2)^2 = 1 + 1 + 4 + 4 = 10$
- $S_{12} = 1(-1) + (-1)(1) + 2(2) + (-2)(-2) = -1 - 1 + 4 + 4 = 6$
- $S_{21} = 6$ (Symmetric)
- $S_{22} = (-1)^2 + 1^2 + 2^2 + (-2)^2 = 10$

$$S = \begin{pmatrix} 10 & 6 \\ 6 & 10 \end{pmatrix}$$

Step 3: Eigenvalues and Eigenvectors Find eigenvalues λ by solving $\det(S - \lambda I) = 0$:

$$(10 - \lambda)(10 - \lambda) - 36 = 0$$

$$(10 - \lambda)^2 = 36 \implies 10 - \lambda = \pm 6$$

- $\lambda_1 = 16$
- $\lambda_2 = 4$

Find eigenvectors:

- **For $\lambda_1 = 16$:**

$$\begin{pmatrix} 10-16 & 6 \\ 6 & 10-16 \end{pmatrix} \mathbf{v} = \begin{pmatrix} -6 & 6 \\ 6 & -6 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \mathbf{0}$$

$$-6v_1 + 6v_2 = 0 \implies v_1 = v_2. \text{ Eigenvector } \mathbf{v}_1 = (1, 1). \text{ Unit Vector: } \mathbf{u}_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right).$$

- For $\lambda_2 = 4$:

$$\begin{pmatrix} 6 & 6 \\ 6 & 6 \end{pmatrix} \mathbf{v} = \mathbf{0}$$

$$6v_1 + 6v_2 = 0 \implies v_1 = -v_2. \text{ Eigenvector } \mathbf{v}_2 = (-1, 1). \text{ Unit Vector: } \mathbf{u}_2 = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right).$$

PCA Choice: The PCA algorithm chooses the direction with the **largest variance** (largest eigenvalue). **Selected Principal Component:** $\mathbf{u}_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right)$.

Step 4: Projections onto the Principal Direction To find the projections, we project the **centered** points onto $\mathbf{u}_1$ and add the mean back (if we want the coordinate in the original space).

Formula: $P_i = \mu + (\mathbf{b}_i \cdot \mathbf{u}_1)\mathbf{u}_1$

- **Point 1** ($x_1 = (4, 1)$, centered $b_1 = (1, -1)$): Dot product: $(1)\left(\frac{1}{\sqrt{2}}\right) + (-1)\left(\frac{1}{\sqrt{2}}\right) = 0$. Projection is the Mean: **(3, 2)**.
- **Point 2** ($x_2 = (2, 3)$, centered $b_2 = (-1, 1)$): Dot product: $(-1)\left(\frac{1}{\sqrt{2}}\right) + (1)\left(\frac{1}{\sqrt{2}}\right) = 0$. Projection is the Mean: **(3, 2)**.
- **Point 3** ($x_3 = (5, 4)$, centered $b_3 = (2, 2)$): Dot product: $(2)\left(\frac{1}{\sqrt{2}}\right) + (2)\left(\frac{1}{\sqrt{2}}\right) = \frac{4}{\sqrt{2}} = 2\sqrt{2}$. Projection: $(3, 2) + 2\sqrt{2}\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = (3, 2) + (2, 2) = \mathbf{(5, 4)}$.
- **Point 4** ($x_4 = (1, 0)$, centered $b_4 = (-2, -2)$): Dot product: $(-2)\left(\frac{1}{\sqrt{2}}\right) + (-2)\left(\frac{1}{\sqrt{2}}\right) = -2\sqrt{2}$. Projection: $(3, 2) - 2\sqrt{2}\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = (3, 2) - (2, 2) = \mathbf{(1, 0)}$.

Final Projections (Original Coordinates):

1. (3, 2)
2. (3, 2)
3. (5, 4)
4. (1, 0)

7. Weighted Least Squares Estimation

Problem: Suppose you measure your pulse rate x three times and get values

$x = b_1, x = b_2, x = b_3$. Each measurement is independent and has zero mean errors with variances $\sigma_1^2 = 1/16, \sigma_2^2 = 1/9$, and $\sigma_3^2 = 1/36$. Find the best estimate $\hat{x}$.

Solution:

1. Model Identification: This is a **Weighted Least Squares (WLS)** problem because the measurements have different variances (precisions). We are trying to fit the model $x = b_i$, or in

matrix form $A\hat{x} = \mathbf{b} + \epsilon$, where $A = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

2. Determine Weights: The weight w_i for each measurement is the inverse of its variance ($w_i = \frac{1}{\sigma_i^2}$). Higher variance (more uncertainty) means lower weight.

- $w_1 = \frac{1}{1/16} = 16$
- $w_2 = \frac{1}{1/9} = 9$
- $w_3 = \frac{1}{1/36} = 36$

3. Weighted Least Squares Formula: The normal equation for WLS is: $(A^T W A)\hat{x} = A^T W \mathbf{b}$. For a single variable scalar estimation, this simplifies to the weighted average:

$$\hat{x} = \frac{\sum w_i b_i}{\sum w_i}$$

4. Calculation:

- Numerator: $w_1 b_1 + w_2 b_2 + w_3 b_3 = 16b_1 + 9b_2 + 36b_3$
- Denominator: $w_1 + w_2 + w_3 = 16 + 9 + 36 = 61$

Answer: The best estimate $\hat{x}$ is:

$$\hat{x} = \frac{16b_1 + 9b_2 + 36b_3}{61}$$

8. Limitations of Gaussian Mixture Models (GMM)

1. Sensitivity to Initialization: GMMs are typically trained using the **Expectation-Maximization (EM)** algorithm, which is an iterative optimization method. The objective function (log-likelihood) is **non-convex**, meaning it has many local maxima.

- If the initial guess for the means, covariances, or weights is poor, the algorithm may get stuck in a **local optimum** rather than finding the global best fit.
- Different initializations can lead to very different clustering results (instability).

2. Determining the Number of Components (k): Unlike some non-parametric methods, GMM requires you to specify the number of Gaussian components (clusters), k , **beforehand**.

- If k is chosen incorrectly, the model will either overfit (too many components) or underfit (too few components) the data.
- Finding the optimal k usually requires extra techniques like the Bayesian Information Criterion (BIC) or Akaike Information Criterion (AIC).

3. Assumption of Gaussian Geometry: GMM assumes that each cluster can be modeled by a Gaussian distribution. This implies that the clusters are expected to be **elliptical** (or spherical) in shape.

- The model struggles to capture clusters with complex, non-convex, or irregular geometric shapes (e.g., spirals, crescents).
- While using full covariance matrices allows for rotation and stretching of the ellipses, it cannot model fundamental non-linear manifold structures effectively without using a very large number of components.

9. Limitations of Expectation Maximization (EM) Algorithm

1. Convergence to Local Optima (Initialization Sensitivity): The EM algorithm guarantees that the likelihood increases (or stays the same) with each iteration, but it does **not** guarantee finding the global maximum likelihood.

- It often gets stuck in **local maxima** or saddle points.
- Because of this, the final result is highly sensitive to the initial parameter values. It is standard practice to run the algorithm multiple times with random initializations to find the best solution.

2. Slow Convergence Rate: The convergence rate of the EM algorithm is typically **linear** (first-order).

- While it may make rapid progress initially, it often slows down significantly as it approaches the optimum.
- This makes it computationally expensive for large datasets or complex models where high precision is required.

3. Need for Probabilistic Assumptions: EM is not a "black box" method; it requires you to assume a specific underlying probability distribution for the data (e.g., Gaussian, Poisson, Multinomial).

- If the true distribution of the data fundamentally differs from the assumed model (model misspecification), the parameters estimated by EM will be biased or meaningless.

4. Singularity Issues (Variance Collapse): In the context of mixture models (like GMM), EM can face numerical stability issues.

- If a component "collapses" onto a single data point (or very few points), the estimated variance for that component approaches zero ($\sigma^2 \rightarrow 0$).
- This causes the likelihood function to explode towards infinity (a singularity), causing the algorithm to crash or produce invalid results. Regularization (adding a small value to the diagonal of the covariance matrix) is often required to prevent this.

10. Applications of Kalman Filters

1. Guidance, Navigation, and Control (GNC): This is the classic application (used in the Apollo program). Kalman filters fuse noisy sensor data from GPS, Inertial Measurement Units (IMUs), and accelerometers to estimate the precise position, velocity, and orientation of aircraft, spacecraft, ships, and submarines.

2. Computer Vision (Object Tracking): Kalman filters are used to track moving objects (like vehicles, pedestrians, or balls in sports) across video frames. Even if the object is temporarily occluded or the detection is noisy, the filter predicts the object's future location based on its estimated velocity and previous trajectory.

3. Robotics (Sensor Fusion & SLAM): Robots use Kalman filters (often Extended Kalman Filters, EKF) to estimate their own location within a map (Localization) and to build the map

The Quadratic Form: $x^T Ax$ and its Derivative

Problem Statement

We are interested in evaluating the derivative with respect to vector x of the scalar function:

$$f(x) = x^T Ax$$

where A is a constant matrix.

Part (a): Evaluating $x^T Ax$ in $\mathbb{R}^2$

Given:

- $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$
- $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$

Step 1: Compute the product Ax

$$Ax = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} A_{11}x_1 + A_{12}x_2 \\ A_{21}x_1 + A_{22}x_2 \end{pmatrix}$$

Step 2: Compute $x^T(Ax)$

$$x^T Ax = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} A_{11}x_1 + A_{12}x_2 \\ A_{21}x_1 + A_{22}x_2 \end{pmatrix}$$

Perform the dot product:

$$\begin{aligned} &= x_1(A_{11}x_1 + A_{12}x_2) + x_2(A_{21}x_1 + A_{22}x_2) \\ &= A_{11}x_1^2 + A_{12}x_1x_2 + A_{21}x_2x_1 + A_{22}x_2^2 \end{aligned}$$

Why is it called the Quadratic Form? It is called a "Quadratic Form" because every term in the resulting polynomial is of **degree 2**.

- x_1^2 and x_2^2 are quadratic terms.
- x_1x_2 is a cross-term, which is also degree 2 (degree 1 in x_1 + degree 1 in x_2). In general, a quadratic form is a homogeneous polynomial of degree 2 in multiple variables.

Part (b): Definition of the Derivative

To evaluate $d(x^T Ax)/dx$, we need the definition of the derivative of a **scalar function** with respect to a **vector**. This is known as the **Gradient**.

Definition: If $f(x)$ is a scalar function of a vector $x = [x_1, \dots, x_n]^T$, the derivative (gradient) is defined as the column vector of partial derivatives:

$$\frac{df}{dx} = \nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{pmatrix}$$

Part (c): Evaluate Derivative for $\mathbb{R}^2$

Let $f(x) = A_{11}x_1^2 + A_{12}x_1x_2 + A_{21}x_2x_1 + A_{22}x_2^2$.

Step 1: Partial Derivative w.r.t x_1 Treat x_2 as a constant.

$$\begin{aligned} \frac{\partial f}{\partial x_1} &= \frac{\partial}{\partial x_1}(A_{11}x_1^2) + \frac{\partial}{\partial x_1}(A_{12}x_1x_2) + \frac{\partial}{\partial x_1}(A_{21}x_2x_1) + \frac{\partial}{\partial x_1}(A_{22}x_2^2) \\ &= 2A_{11}x_1 + A_{12}x_2 + A_{21}x_2 + 0 \\ &= 2A_{11}x_1 + (A_{12} + A_{21})x_2 \end{aligned}$$

Step 2: Partial Derivative w.r.t x_2 Treat x_1 as a constant.

$$\begin{aligned} \frac{\partial f}{\partial x_2} &= 0 + A_{12}x_1 + A_{21}x_1 + 2A_{22}x_2 \\ &= (A_{12} + A_{21})x_1 + 2A_{22}x_2 \end{aligned}$$

Step 3: Assemble the Gradient Vector

$$\frac{df}{dx} = \begin{pmatrix} 2A_{11}x_1 + (A_{12} + A_{21})x_2 \\ (A_{12} + A_{21})x_1 + 2A_{22}x_2 \end{pmatrix}$$

We can decompose this vector back into matrix form to see the pattern:

$$\begin{aligned} &= \begin{pmatrix} 2A_{11} & A_{12} + A_{21} \\ A_{12} + A_{21} & 2A_{22} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ &= \left[\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} + \begin{pmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \end{pmatrix} \right] \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \\ &= (A + A^T)x \end{aligned}$$

Answer for (c):

$$\frac{d(x^T A x)}{dx} = (A + A^T)x$$

Part (d): Generalization to $\mathbb{R}^n$

Let $x \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n \times n}$. The scalar function can be written using summation notation:

$$f(x) = \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j$$

Calculate partial derivative w.r.t x_k (where k is a specific index from 1 to n):

$$\frac{\partial f}{\partial x_k} = \frac{\partial}{\partial x_k} \left(\sum_{i=1}^n \sum_{j=1}^n A_{ij} x_i x_j \right)$$

We apply the product rule. The term $x_i x_j$ has a derivative w.r.t x_k only when $i = k$ or $j = k$.

- Case 1: $i = k$ and $j \neq k$** Term is $A_{kj} x_k x_j$. Derivative is $A_{kj} x_j$. Summing over all $j \neq k$:
 $\sum_{j \neq k} A_{kj} x_j$.
- Case 2: $j = k$ and $i \neq k$** Term is $A_{ik} x_i x_k$. Derivative is $A_{ik} x_i$. Summing over all $i \neq k$:
 $\sum_{i \neq k} A_{ik} x_i$.
- Case 3: $i = k$ and $j = k$** Term is $A_{kk} x_k^2$. Derivative is $2A_{kk} x_k$. This can be split into
 $A_{kk} x_k + A_{kk} x_k$.

Combining the sums:

$$\frac{\partial f}{\partial x_k} = \sum_{j=1}^n A_{kj} x_j + \sum_{i=1}^n A_{ik} x_i$$

- The first sum $\sum_j A_{kj} x_j$ corresponds to the k -th element of the vector Ax . (Row k of A dot product with x).
- The second sum $\sum_i A_{ik} x_i$ corresponds to the k -th element of the vector $A^T x$. (Row k of A^T dot product with x , which is Col k of A).

So, the gradient vector containing all k components is:

$$\nabla f(x) = Ax + A^T x$$

Matrix Form Result:

$$\frac{d(x^T A x)}{dx} = (A + A^T)x$$

Part (e): Case where A is Symmetric

If A is a symmetric matrix, then by definition:

$$A^T = A$$

Substituting this into our general result from (d):

$$\frac{d(x^T Ax)}{dx} = (A + A)x = 2Ax$$

Comparison to Scalar Calculus: This result is beautifully analogous to standard calculus.

- Scalar: $f(x) = ax^2 \implies f'(x) = 2ax$.
- Vector: $f(x) = x^T Ax \implies \nabla f(x) = 2Ax$. (If A is symmetric).

Vector Calculus: The Chain Rule for Composite Functions

Problem Statement

We are given two vector-valued functions:

1. **Inner Function (f):** A function from $\mathbb{R}^3 \rightarrow \mathbb{R}^2$.

$$f(x) = \begin{pmatrix} x_1^2 + x_2^2 + x_3^2 \\ 2x_1 - x_2 - x_3 \end{pmatrix}$$

Let $z = f(x)$, where $z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}$.

2. **Outer Function (g):** A function from $\mathbb{R}^2 \rightarrow \mathbb{R}^2$.

$$g(z) = \begin{pmatrix} z_2/z_1 \\ z_1 z_2 \end{pmatrix}$$

Let $y(x) = g(f(x))$.

Goal: Compute the Jacobian matrix of the composite function y with respect to x :

$$\frac{\partial y}{\partial x^T} = J_y(x)$$

1. The Chain Rule Formula

For a composite function $y = g(f(x))$, the Jacobian matrix is the product of the Jacobian of the outer function and the Jacobian of the inner function.

$$\frac{\partial y}{\partial x^T} = \frac{\partial g}{\partial z^T} \cdot \frac{\partial f}{\partial x^T}$$

Dimensions Check:

- y is 2×1 . x is 3×1 . The result $\frac{\partial y}{\partial x^T}$ must be 2×3 .
- $\frac{\partial g}{\partial z^T}$ is the Jacobian of g w.r.t z . Dimensions: 2×2 .
- $\frac{\partial f}{\partial x^T}$ is the Jacobian of f w.r.t x . Dimensions: 2×3 .
- Product: $(2 \times 2) \cdot (2 \times 3) = 2 \times 3$. The dimensions match.

2. Step 1: Compute Jacobian of Inner Function (J_f)

Let $z = f(x)$.

$$z_1 = x_1^2 + x_2^2 + x_3^2$$

$$z_2 = 2x_1 - x_2 - x_3$$

The Jacobian $J_f = \begin{pmatrix} \nabla z_1^T \\ \nabla z_2^T \end{pmatrix}$:

$$J_f = \begin{pmatrix} \frac{\partial z_1}{\partial x_1} & \frac{\partial z_1}{\partial x_2} & \frac{\partial z_1}{\partial x_3} \\ \frac{\partial z_2}{\partial x_1} & \frac{\partial z_2}{\partial x_2} & \frac{\partial z_2}{\partial x_3} \end{pmatrix}$$

Derivatives for Row 1 (z_1):

- $\frac{\partial}{\partial x_1}(x_1^2 + \dots) = 2x_1$
- $\frac{\partial}{\partial x_2}(\dots + x_2^2 + \dots) = 2x_2$
- $\frac{\partial}{\partial x_3}(\dots + x_3^2) = 2x_3$

Derivatives for Row 2 (z_2):

- $\frac{\partial}{\partial x_1}(2x_1 - \dots) = 2$
- $\frac{\partial}{\partial x_2}(\dots - x_2 - \dots) = -1$
- $\frac{\partial}{\partial x_3}(\dots - x_3) = -1$

Resulting Matrix J_f :

$$J_f = \begin{pmatrix} 2x_1 & 2x_2 & 2x_3 \\ 2 & -1 & -1 \end{pmatrix}$$

3. Step 2: Compute Jacobian of Outer Function (J_g)

Let $y = g(z)$.

$$y_1 = \frac{z_2}{z_1} = z_2 z_1^{-1}$$

$$y_2 = z_1 z_2$$

The Jacobian $J_g = \begin{pmatrix} \nabla y_1^T \\ \nabla y_2^T \end{pmatrix}$:

$$J_g = \begin{pmatrix} \frac{\partial y_1}{\partial z_1} & \frac{\partial y_1}{\partial z_2} \\ \frac{\partial y_2}{\partial z_1} & \frac{\partial y_2}{\partial z_2} \end{pmatrix}$$

Derivatives for Row 1 ($y_1 = z_2/z_1$):

- $\frac{\partial}{\partial z_1}(z_2 z_1^{-1}) = -z_2 z_1^{-2} = -\frac{z_2}{z_1^2}$
- $\frac{\partial}{\partial z_2}(z_2 z_1^{-1}) = \frac{1}{z_1}$

Derivatives for Row 2 ($y_2 = z_1 z_2$):

- $\frac{\partial}{\partial z_1}(z_1 z_2) = z_2$
- $\frac{\partial}{\partial z_2}(z_1 z_2) = z_1$

Resulting Matrix J_g :

$$J_g = \begin{pmatrix} -\frac{z_2}{z_1^2} & \frac{1}{z_1} \\ z_2 & z_1 \end{pmatrix}$$

4. Step 3: Matrix Multiplication (The Chain Rule)

Now we multiply $J_g \cdot J_f$.

$$J_y = \begin{pmatrix} -\frac{z_2}{z_1^2} & \frac{1}{z_1} \\ z_2 & z_1 \end{pmatrix} \begin{pmatrix} 2x_1 & 2x_2 & 2x_3 \\ 2 & -1 & -1 \end{pmatrix}$$

We calculate this row by row.

Row 1 of Final Matrix:

- **Col 1:** $(-\frac{z_2}{z_1^2})(2x_1) + (\frac{1}{z_1})(2) = \frac{2}{z_1} - \frac{2x_1 z_2}{z_1^2} = \frac{2z_1 - 2x_1 z_2}{z_1^2}$
- **Col 2:** $(-\frac{z_2}{z_1^2})(2x_2) + (\frac{1}{z_1})(-1) = -\frac{1}{z_1} - \frac{2x_2 z_2}{z_1^2} = -\frac{z_1 + 2x_2 z_2}{z_1^2}$
- **Col 3:** $(-\frac{z_2}{z_1^2})(2x_3) + (\frac{1}{z_1})(-1) = -\frac{1}{z_1} - \frac{2x_3 z_2}{z_1^2} = -\frac{z_1 + 2x_3 z_2}{z_1^2}$

Row 2 of Final Matrix:

- **Col 1:** $(z_2)(2x_1) + (z_1)(2) = 2(x_1 z_2 + z_1)$
- **Col 2:** $(z_2)(2x_2) + (z_1)(-1) = 2x_2 z_2 - z_1$
- **Col 3:** $(z_2)(2x_3) + (z_1)(-1) = 2x_3 z_2 - z_1$

5. Final Result

The Jacobian matrix $\frac{\partial y}{\partial x^T}$ expressed in terms of x and intermediate variables z_1, z_2 is:

$$\frac{\partial y}{\partial x^T} = \begin{pmatrix} \frac{2(z_1 - x_1 z_2)}{z_1^2} & -\frac{z_1 + 2x_2 z_2}{z_1^2} & -\frac{z_1 + 2x_3 z_2}{z_1^2} \\ 2(x_1 z_2 + z_1) & 2x_2 z_2 - z_1 & 2x_3 z_2 - z_1 \end{pmatrix}$$

Where:

- $z_1 = x_1^2 + x_2^2 + x_3^2$
- $z_2 = 2x_1 - x_2 - x_3$

Optimization & Least Squares Problems

13. Stationary Point Analysis (6 Marks)

Problem: The function $f(x, y, z) = x^2 + y^2 + 3z^2 - xy + 2xz + yz$ defined on $\mathbb{R}^3$ has only one stationary point. Show that it is a local minimum.

Solution:

Step 1: Find the Gradient Vector (∇f) To find the stationary points, we first compute the partial derivatives with respect to x, y , and z .

$$\nabla f = \begin{pmatrix} \frac{\partial f}{\partial x} \\ \frac{\partial f}{\partial y} \\ \frac{\partial f}{\partial z} \end{pmatrix} = \begin{pmatrix} 2x - y + 2z \\ 2y - x + z \\ 6z + 2x + y \end{pmatrix}$$

Step 2: Solve $\nabla f = \mathbf{0}$ for Stationary Points We set the gradient to the zero vector to obtain a system of linear equations:

1. $2x - y + 2z = 0$
2. $-x + 2y + z = 0$
3. $2x + y + 6z = 0$

Solving the system:

- From (1), we can express y as: $y = 2x + 2z$.
- Substitute this into (2):

$$-x + 2(2x + 2z) + z = 0$$

$$-x + 4x + 4z + z = 0$$

$$3x + 5z = 0 \implies x = -\frac{5}{3}z$$

- Now substitute x back into the expression for y :

$$y = 2\left(-\frac{5}{3}z\right) + 2z = -\frac{10}{3}z + \frac{6}{3}z = -\frac{4}{3}z$$

- Substitute both x and y into equation (3) to verify and solve for z :

$$2\left(-\frac{5}{3}z\right) + \left(-\frac{4}{3}z\right) + 6z = 0$$

$$-\frac{10}{3}z - \frac{4}{3}z + \frac{18}{3}z = 0$$

$$\frac{4}{3}z = 0 \implies z = 0$$

Since $z = 0$, it follows that $x = 0$ and $y = 0$. **Conclusion:** The only stationary point is the origin $(0, 0, 0)$.

Step 3: Analyze the Hessian Matrix To classify the stationary point, we compute the Hessian Matrix (the matrix of second derivatives).

$$H_f = \begin{pmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{pmatrix} = \begin{pmatrix} 2 & -1 & 2 \\ -1 & 2 & 1 \\ 2 & 1 & 6 \end{pmatrix}$$

Step 4: Check for Positive Definiteness (Sylvester's Criterion) A stationary point is a **local minimum** if the Hessian matrix is **Positive Definite**. We check this by calculating the leading principal minors (D_i).

1. **First Minor (D_1):**

$$D_1 = \det[2] = 2 > 0$$

2. **Second Minor (D_2):**

$$D_2 = \det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = (2)(2) - (-1)(-1) = 4 - 1 = 3 > 0$$

3. **Third Minor (D_3 - The Determinant of H):**

$$D_3 = 2 \det \begin{pmatrix} 2 & 1 \\ 1 & 6 \end{pmatrix} - (-1) \det \begin{pmatrix} -1 & 1 \\ 2 & 6 \end{pmatrix} + 2 \det \begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix}$$

$$= 2(12 - 1) + 1(-6 - 2) + 2(-1 - 4)$$

$$= 2(11) + 1(-8) + 2(-5)$$

$$= 22 - 8 - 10$$

$$D_3 = 4 > 0$$

Final Conclusion: Since all leading principal minors are strictly positive ($D_1 > 0, D_2 > 0, D_3 > 0$), the Hessian matrix is **Positive Definite**. Therefore, the stationary point $(0, 0, 0)$ is a **Local Minimum**.

14. Least Squares Solution (10 Marks)

Problem: Find the least square solution to the following system of equations:

$$\begin{aligned}x_1 + x_2 &= 2 \\x_1 - x_2 &= 0 \\2x_1 + x_2 &= -4\end{aligned}$$

Solution:

Step 1: Matrix Formulation ($A\mathbf{x} = \mathbf{b}$) We write the system in matrix form. Since there are 3 equations and 2 unknowns, the system is overdetermined ($m > n$).

$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ 0 \\ -4 \end{pmatrix}$$

Step 2: The Normal Equation Since $A\mathbf{x} = \mathbf{b}$ has no exact solution (the vector $\mathbf{b}$ is likely not in the column space of A), we solve the **Normal Equation** for the least squares estimate $\hat{\mathbf{x}}$:

$$A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$$

Step 3: Compute $A^T A$

$$\begin{aligned}A^T A &= \begin{pmatrix} 1 & 1 & 2 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 1 \end{pmatrix} \\&= \begin{pmatrix} (1)(1) + (1)(1) + (2)(2) & (1)(1) + (1)(-1) + (2)(1) \\ (1)(1) + (-1)(1) + (1)(2) & (1)(1) + (-1)(-1) + (1)(1) \end{pmatrix} \\&= \begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix}\end{aligned}$$

Step 4: Compute $A^T \mathbf{b}$

$$\begin{aligned}A^T \mathbf{b} &= \begin{pmatrix} 1 & 1 & 2 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 0 \\ -4 \end{pmatrix} \\&= \begin{pmatrix} (1)(2) + (1)(0) + (2)(-4) \\ (1)(2) + (-1)(0) + (1)(-4) \end{pmatrix} \\&= \begin{pmatrix} 2 - 8 \\ 2 - 4 \end{pmatrix} = \begin{pmatrix} -6 \\ -2 \end{pmatrix}\end{aligned}$$

Step 5: Solve for $\hat{\mathbf{x}}$ We now have the system:

$$\begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} \hat{x}_1 \\ \hat{x}_2 \end{pmatrix} = \begin{pmatrix} -6 \\ -2 \end{pmatrix}$$

We can use row reduction (Gaussian Elimination) or Cramer's rule. Let's use substitution from the second equation:

1. $6\hat{x}_1 + 2\hat{x}_2 = -6 \implies 3\hat{x}_1 + \hat{x}_2 = -3 \implies \hat{x}_2 = -3 - 3\hat{x}_1$
2. Substitute into the second row equation ($2\hat{x}_1 + 3\hat{x}_2 = -2$):

$$2\hat{x}_1 + 3(-3 - 3\hat{x}_1) = -2$$

$$2\hat{x}_1 - 9 - 9\hat{x}_1 = -2$$

$$-7\hat{x}_1 = 7$$

$$\hat{x}_1 = -1$$

3. Find $\hat{x}_2$:

$$\hat{x}_2 = -3 - 3(-1) = -3 + 3 = 0$$

Final Answer: The least squares solution is:

$$\hat{\mathbf{x}} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

Lagrange Multipliers: Optimization with Constraints

15. Minimization Problem (16 Marks)

Problem Statement: Use the method of Lagrange multipliers to find the minimum value of the function $f(x, y) = x^2 + 4y^2 - 2x + 8y$ subject to the constraint $x + 2y = 7$.

Solution:

Step 1: Define the Objective and Constraint Functions

- **Objective Function (f):** This is the function we want to minimize.

$$f(x, y) = x^2 + 4y^2 - 2x + 8y$$

- **Constraint Function (g):** This is the condition that must be satisfied. We write it in the form $g(x, y) = c$.

$$g(x, y) = x + 2y = 7$$

$$g(x, y) - 7 = 0$$

Step 2: Formulate the Lagrangian Function

The method of Lagrange multipliers relies on the auxiliary function $L(x, y, \lambda)$:

$$L(x, y, \lambda) = f(x, y) - \lambda(g(x, y) - c)$$

Substituting our functions:

$$L(x, y, \lambda) = (x^2 + 4y^2 - 2x + 8y) - \lambda(x + 2y - 7)$$

Step 3: Compute Partial Derivatives

To find the critical points, we need the partial derivatives of L with respect to x , y , and λ to be zero.

1. **Derivative with respect to x :**

$$\frac{\partial L}{\partial x} = \frac{\partial}{\partial x}(x^2 - 2x) - \frac{\partial}{\partial x}(\lambda x)$$

$$\frac{\partial L}{\partial x} = 2x - 2 - \lambda$$

2. **Derivative with respect to y :**

$$\frac{\partial L}{\partial y} = \frac{\partial}{\partial y}(4y^2 + 8y) - \frac{\partial}{\partial y}(\lambda 2y)$$

$$\frac{\partial L}{\partial y} = 8y + 8 - 2\lambda$$

3. Derivative with respect to λ :

$$\frac{\partial L}{\partial \lambda} = -(x + 2y - 7)$$

Step 4: Set up the System of Equations

Setting the derivatives to zero gives us the following system:

- **Equation (1):** $2x - 2 - \lambda = 0 \implies \lambda = 2x - 2$
- **Equation (2):** $8y + 8 - 2\lambda = 0 \implies 2\lambda = 8y + 8$
- **Equation (3):** $x + 2y = 7$ (The Constraint)

Step 5: Solve the System

We need to solve for x and y .

1. Express λ in terms of y from Eq (2):

$$2\lambda = 8y + 8$$

Divide by 2:

$$\lambda = 4y + 4$$

2. Equate the two expressions for λ : From Eq (1) and the result above:

$$2x - 2 = 4y + 4$$

3. Simplify to find a relationship between x and y :

$$2x - 4y = 6$$

Divide by 2:

$$x - 2y = 3 \implies x = 2y + 3$$

4. Substitute x into the Constraint Equation (Eq 3):

$$(2y + 3) + 2y = 7$$

$$4y + 3 = 7$$

$$4y = 4$$

$$y = 1$$

5. **Find x :** Substitute $y = 1$ back into the expression for x :

$$x = 2(1) + 3$$

$$x = 5$$

Stationary Point: $(x, y) = (5, 1)$.

Step 6: Calculate the Minimum Value

Now, substitute the point $(5, 1)$ back into the original objective function $f(x, y)$.

$$f(5, 1) = (5)^2 + 4(1)^2 - 2(5) + 8(1)$$

$$f(5, 1) = 25 + 4 - 10 + 8$$

$$f(5, 1) = 29 - 10 + 8$$

$$f(5, 1) = 19 + 8$$

$$f(5, 1) = 27$$

Step 7: Verification (Optional but Recommended for Full Marks)

Since the objective function

$f(x, y) = (x - 1)^2 - 1 + (2y + 2)^2 - 4 = (x - 1)^2 + 4(y + 1)^2 - 5$ represents an elliptic paraboloid opening upwards, any stationary point found under a linear constraint must be a **minimum**.

Final Answer: The minimum value of the function subject to the constraint is **27**, occurring at the point **(5, 1)**.